COLOMBIA

Short-Term Rental Market Research System

Engineering Execution Plan — Powered by AirROI API

Bogota · Cali (Granada, El Peñon, Calima, Pance) · Salento · Cartagena · Medellin · Santa Marta

Item	Detail
Pre-approval budget	COP $1,766,000,000 (≈ USD $425,600 at COP 4,150/USD)
Target markets	Bogota (Chapinero, Usaquen) · Cali (Granada, El Peñon, Calima, Pance) · Salento · Cartagena · Medellin · Santa Marta
Primary STR data	AirROI API — 22 REST endpoints, $0.01/call, no contract, instant key
Property prices	Finca Raiz scraper (COP asking prices)
Backend	Python 3.11+ + FastAPI + Pydantic v2
Frontend	React 18 + TypeScript + Vite
Database	PostgreSQL 16 + PostGIS extension
Hosting (MVP)	Railway (backend + DB + cron) + Vercel (frontend)
Est. monthly running cost	~$30–$75 USD at MVP scale vs $300–$500/month for AirDNA

Part 1 — Budget Analysis & Market Fit

Your pre-approval of COP $1,766,000,000 (≈ USD $425,600) is a substantial budget that opens quality STR-grade properties across most Colombian markets. However, purchasing power varies significantly by city — it buys 1 apartment in Cartagena or 4–5 fincas in Calima. The table below ranks all 8 markets by investment fit, defined as the combination of price-to-revenue ratio, how many properties your budget can purchase, occupancy reliability, and regulatory risk.

Cali market expanded: This version expands the Cali analysis to include Calima (Lake Calima / Daríen) and Pance — two rural sub-markets within Valle del Cauca that offer meaningfully higher ADR and lower purchase prices than urban Cali apartments, with no HOA restrictions on standalone properties.

Market	ADR (USD)	Occupancy	Est. annual revenue	Avg. price range (COP)	Units at budget	Fit & rationale
Cali region — expanded (ranked by investment fit)						
Calima (Lake Calima)	$65–$120	55–70%	~$16K–$22K	COP 280–500M	**3–5 units**	**Strong fit**

Market	ADR (USD)	Occupancy	Est. annual revenue	Avg. price range (COP)	Units at budget	Fit & rationale
Valle del Cauca — lake & watersports				Fincas & lake houses	Highest portfolio count	Highest ADR vs. price in the region. Watersports tourism (kitesurfing, windsurfing) drives consistent weekend demand. Lowest prices = most units per peso. Emerging — low competition.
Pance Southern Cali — nature retreat	$60–$95	55–65%	~$13.5K–$18K	COP 450–700M Cabins & retreats	**2–3 units** City + nature combo	**Strong fit** 30 min from Cali center. Luxury cabins with Farallones views command premium rates. No HOA or building committee restrictions on standalone properties.
Salento Quindio — Coffee Region	$55–$75	60–70%	~$14.6K–$19.2K	COP 350–600M Fincas & townhouses	**3–4 units** Best occupancy rate	**Strong fit** Coffee Region tourism drives year-round international demand. Highest sustained occupancy of all markets. No HOA on most finca-style properties. Minimal regulatory risk.
Cali — Granada & El Peñon Urban Cali — nightlife & culture	$30–$46	40–53%	~$7.6K–$9.2K	COP 500–750M STR apartments	**2–3 units** Urban diversification	**Good fit** Lower ADR but strong urban demand. World Salsa Festival (June) spikes rates 30–50%. Less saturated than Bogota or Medellin. Gross yields 6–8% achievable.
Other target markets						
Bogota Chapinero & Usaquen	$34–$73	53–60%	~$9.2K–$14.6K	COP 700M–1.1B STR apartments	**1–2 units** Capital city liquidity	**Good fit** Largest STR market in Colombia (22,000+ listings). Year-round corporate + tourism demand. HOA COP 400K–1.2M/month is a significant cost. RNT enforcement tightening in 2026.
Medellin Laureles & El Poblado	$33–$85	55–65%	~$10.8K–$15.6K	COP 750M–1.1B STR apartments	**1–2 units** Highest ADR ceiling	**Good fit** Best occupancy of urban markets (63% median). El Poblado is heavily saturated — top performers drive the averages. High regulation (88% licensed). Budget buys 1–2 units only.
Santa Marta Magdalena — Caribbean beach	$44–$89	32–45%	~$7.9K–$11K	COP 500–800M Beach apartments	**2 units** High seasonal risk	**Moderate fit** High ADR ceiling but lowest occupancy (32% average). Supply grew 114% YoY — very competitive. Highly seasonal: strong Jan–Mar, very soft May–Sep. Best as a secondary market.
Cartagena Getsemani & Centro Hist.	$65–$96+	47–57%	~$18,036	COP 1.0–1.4B Premium STR units	**1 unit** Zero diversification	**Use caution** Highest revenue per property but budget buys exactly 1 unit with no reserve. If that unit has vacancy or maintenance issues, all income stops. Consider after building cash flow elsewhere.

Note: Est. annual revenue = median ADR × projected occupancy × 365. Price ranges reflect STR-grade properties in prime STR sub-neighborhoods. “Units at budget” assumes all-cash purchase at price range midpoint. Data sourced from AirROI, TheLatinvestor, and GlobalPropertyGuide (2025–2026).

1.1 Recommended portfolio strategies

Strategy	Allocation	Est. properties	Why it works
Diversified — rural + nature	2 Calima + 2 Salento + 1 Pance	5 properties (~COP 1.6B)	Maximum portfolio count. All 3 markets have strong occupancy, low purchase prices, and zero HOA costs. Geographic spread reduces risk. Best use of the full budget.
Balanced — rural + urban	1 Bogota + 1 Salento + 1 Pance	3 properties (~COP 1.65B)	Mixes stable year-round urban income (Bogota corporate demand) with high-occupancy rural units. Diversifies guest types and seasonal risk.
Concentrated — Cartagena only	1 prime Cartagena unit	1 property (~COP 1.2–1.4B)	Highest single-property revenue ceiling but zero diversification. Recommended only after validating operations in lower-risk markets first.

> **Recommendation:** Start with Calima or Salento. Both markets give you 3+ properties for your budget, have the strongest occupancy-to-price ratios, and carry the lowest regulatory risk. Use the research system to validate 3–5 specific listings before committing capital.

Part 2 — How Each Metric Is Sourced and Calculated

Every metric surfaces from one of three sources: the AirROI API, the Finca Raiz scraper, or a formula your calculation engine computes from those inputs. This table defines the exact method for each of the 9 metrics the system produces.

Metric	Data source	AirROI endpoint	How it's calculated
Avg Daily Rate (ADR)	AirROI	/markets/summary/listings/search/radius	Median booked nightly rate for comparable listings in target area, filtered by bedroom count and property type. Use trailing 12-month window.
Projected Occupancy	AirROI	/markets/occupancy/listings/metrics/all	Booked nights ÷ available nights from listing calendar data, averaged across comparable properties over trailing 12 months.
Projected Annual Revenue	AirROI + Calc	/markets/revenueRevenue Calculator	ADR × Projected Occupancy × 365. AirROI's Revenue Calculator endpoint returns this directly per address or market. Cross-check with the manual formula and surface both figures.
Peak Months	AirROI	/markets/seasonality	Months where ADR or occupancy exceeds the annual average by 15%+. AirROI returns monthly breakdowns. Flag top 3 months. Feeds the seasonal forecast calendar view.
Purchase Price	Finca Raiz scraper	fincaraiz.com.co listing pages	Scraped from the Finca Raiz listing URL (asking price in COP). Also accepts manual entry for off-market properties. Converted to USD at daily exchange rate.
Monthly Expenses	Manual input + formula	No API — user-defined	Fixed: mortgage payment (from purchase price, down %, rate, term) + predial + HOA. Variable: property management fee (18–25% of revenue) + maintenance reserve (1% of purchase price ÷ 12).

Metric	Data source	AirROI endpoint	How it's calculated
Cash on Cash Return	Calc engine	Derived metric	Annual Net Income ÷ Total Cash Invested. Net Income = Annual Revenue − Annual Expenses. Cash Invested = down payment + closing costs (2.5–3%) + renovation budget.
Payback Period	Calc engine	Derived metric	Total Cash Invested ÷ Annual Net Income. Returned in years as a float. Flag any property above 12 years as low priority.
Comparable Listings	AirROI	/listings/comparables	Pass target property lat/lng + bedrooms + property type. Returns 10–20 nearest similar active listings with ADR, occupancy, revenue, and distance. Displayed as a sortable table.

Part 3 — Phased Execution Plan

Four phases, each ending with a working deliverable. The engineer starts today on Phase 1 and should have live market data in the database by end of day.

PHASE 1 AirROI Integration & Database Setup

Duration: Day 1–3 | Deliverable: Verified STR market data for all 8 target markets flowing into PostgreSQL.

Step 1 Sign up for AirROI and validate coverage for all 8 markets

- Go to airroi.com, create a free account, and add $10 in API credits (instant activation).
- Retrieve the API key from the developer dashboard.
- Run a test call to /markets/search for: Bogota, Cali, Salento, Cartagena, Medellin, Santa Marta, Calima (try "Calima, Valle del Cauca" and "Lago Calima"), and Pance.
- For each market: record whether ADR, occupancy, and revenue fields are populated and how many active listings are returned.
- Calima and Pance may have thin coverage in AirROI as standalone markets — if fewer than 30 listings return, fall back to searching the broader "Darien, Valle del Cauca" or "Sur de Cali" market and filtering by proximity.
- Save the raw JSON response for each market to a local file for schema reference before writing any database code.

Step 2 Set up PostgreSQL 16 with PostGIS

- Install PostgreSQL 16 locally or spin up a free instance on Railway or Neon.tech.
- Enable the PostGIS extension: CREATE EXTENSION postgis;
- markets table: id, city, neighborhood, country, adr_usd, occupancy_rate, annual_revenue_usd, peak_months TEXT[], listing_count, last_updated.
- listings table: airroi_id, market_id FK, lat, lng, location GEOGRAPHY(Point,4326), bedrooms, property_type, adr_usd, occupancy_rate, annual_revenue_usd, source_url.

properties table: id, address, lat, lng, bedrooms, property_type, purchase_price_cop BIGINT, purchase_price_usd NUMERIC, source_url, notes, created_at.
property_financial_reports table: id, property_id FK, down_payment_pct, interest_rate, loan_term_years, monthly_expenses_usd, annual_revenue_usd, annual_net_income_usd, cash_invested_usd, coc_return_pct, payback_years NUMERIC, exchange_rate, calculated_at.
exchange_rates table: date DATE PRIMARY KEY, cop_per_usd NUMERIC. Populated daily from a free API.
Create a GiST index on listings.location to support fast radius queries: CREATE INDEX idx_listings_location ON listings USING GIST(location);

Step 3 **Build AirROI data ingestion scripts (Python)**
ingest_markets.py: Calls /markets/summary for all 8 target markets. Writes ADR, occupancy, revenue, and peak months to the markets table. Run once now; schedule nightly.
ingest_listings.py: Calls /listings/search/market for each market with pagination. Writes results to the listings table. Filter for 1–3 bedroom apartments, houses, and fincas.
ingest_exchange_rate.py: Fetches COP/USD rate from frankfurter.app (free, no API key). Writes to exchange_rates. Schedule daily at 9am Colombia time (UTC-5).
Run all three scripts and confirm data is present in the database before moving to Phase 2.

PHASE 2 Finca Raiz Scraper & Calculation Engine

Duration: Day 3–6 | Deliverable: Any candidate property address returns a complete financial projection in JSON.

Step 4 **Build the Finca Raiz price scraper**
Write a lightweight Python scraper using httpx + BeautifulSoup targeting fincaraiz.com.co.
Input: a Finca Raiz listing URL. Output: asking price in COP, neighborhood, bedrooms, property type.
Apply the daily COP/USD rate from the exchange_rates table to convert and store both values in the properties table.
This scraper runs on-demand only when a user adds a new property — not on a schedule.
Also implement a manual entry fallback: accept price directly as a POST body parameter for off-market or private sale properties.
For Calima and Pance properties, also check fincainca.com and metrocuadrado.com as alternative listing sources if Finca Raiz coverage is thin.

Step 5 **Build the financial calculation module (calculations.py)**
mortgage_payment(purchase_price_usd, down_pct, annual_rate, term_years): standard amortization formula, returns monthly payment in USD.
monthly_expenses(mortgage, hoa_usd, mgmt_fee_pct, annual_revenue_usd, property_value_usd): sums mortgage + HOA + management fee + maintenance reserve (1% of value ÷ 12).

annual_net_income(annual_revenue_usd, monthly_expenses_usd): annual revenue minus annualized expenses.
coc_return(annual_net_income_usd, cash_invested_usd): returns percentage.
payback_period(cash_invested_usd, annual_net_income_usd): returns years as a float.
closing_costs(purchase_price_usd): returns 2.75% as the default Colombia estimate (notary + registration + taxes).
All functions take explicit parameters only — no global state. Unit-test each with known inputs before connecting to the API.
Colombian defaults to pre-load in the UI: 10% interest rate, 30% down, 15-year term, 22% management fee, COP 0 HOA for fincas, COP 350,000 HOA for urban apartments.

Step 6 **Wire AirROI revenue data into the calculation engine**
For any candidate property: (1) call /listings/comparables with the property's lat/lng + bedrooms + property type, (2) compute median ADR and median occupancy across the comp set, (3) calculate projected annual revenue as median_adr × median_occupancy × 365.
Also call AirROI's Revenue Calculator endpoint and surface both figures — the direct AirROI estimate and the comp-derived estimate. Mark the lower of the two as the "conservative" figure.
Write the full result to the financials table with a calculated_at timestamp.
Write the comp listings used to a property_comps table linking property_id to listing airroi_id, so comps can be retrieved later without a repeat API call.

PHASE 3 FastAPI Backend

Duration: Day 6–9 | Deliverable: A fully documented REST API serving all market and property data. Testable via Swagger UI at /docs.

Step 7 **Project structure and dependencies**
Install: pip install fastapi uvicorn asyncpg sqlalchemy pydantic httpx beautifulsoup4 python-dotenv
Project layout: /app/main.py, /app/routers/ (markets.py, properties.py, comps.py), /app/services/ (airroi.py, calculations.py, finca_raiz.py, exchange_rate.py), /app/models/ (database.py, schemas.py).
Define all Pydantic v2 response models in schemas.py matching exactly what each endpoint returns. Strict TypeScript types on the frontend are derived from these.
Store AirROI API key, database URL, and exchange rate API URL in a .env file. Never hardcode credentials.

Step 8 **Define all API endpoints**
GET /markets — returns all 8 markets with ADR, occupancy, annual revenue, peak months, listing count, and last_updated.
GET /markets/{city}/listings — returns top 50 listings for a city. Accepts bedrooms and property_type query params. Sortable by revenue or occupancy.

POST /properties — accepts Finca Raiz URL or manual inputs. Triggers the scraper. Returns the saved property object with both COP and USD price.
POST /properties/{id}/analyze — accepts expense inputs (down_payment_pct, interest_rate, hoa_cop, mgmt_fee_pct, renovation_cop). Runs the calculation engine. Writes and returns the financials JSON.
GET /properties/{id}/comps — calls AirROI /listings/comparables. Returns the comp table with 10–20 listings including ADR, occupancy, revenue, and distance in km.
GET /properties/{id}/comps/cached — returns comps from the property_comps table without a new API call (free after first analysis).
GET /exchange-rate — returns the current COP/USD rate and the date it was fetched.
All amounts stored and returned in both COP and USD. All Pydantic response models explicitly typed.

Step 9 **Validate end-to-end with a real Salento or Calima property**
Find a live Finca Raiz listing in Salento or Calima.
POST the URL to /properties — confirm the scraper returns the correct price in COP.
POST to /properties/{id}/analyze with Colombian default expense inputs.
Verify the JSON contains all 9 metrics: ADR (both estimate methods), occupancy, projected revenue, peak months, purchase price (COP + USD), monthly expenses, CoC return, payback period.
GET /properties/{id}/comps — confirm at least 5 comparable listings are returned with distances.
Fix any data quality or schema issues before building the frontend.

PHASE 4 React + TypeScript Dashboard

Duration: Day 9–14 | Deliverable: A deployed React dashboard for evaluating all 8 markets and individual properties.

Step 10 **Project setup (React + TypeScript)**
Scaffold: npm create vite@latest colombia-str -- --template react-ts
Install: react-query (data fetching + caching), react-router-dom v6, axios, recharts (charts), react-leaflet + leaflet (Colombia map), @types/leaflet.
Create /src/types/api.ts with TypeScript interfaces matching every FastAPI Pydantic schema. Use these types throughout — never use "any".
Configure a .env file with VITE_API_URL pointing to the FastAPI backend.
Set up react-query QueryClient with 1-hour stale time for market data and 30-minute stale time for property comps.

Step 11 **Market overview screen**
Route: / (home)
React-Leaflet map of Colombia centered on the 8 target markets. Click a city marker to scroll to its card.

Grid of 8 market cards below the map. Each card shows: city, ADR (USD), projected occupancy %, projected annual revenue (USD), peak months as colored pills, active listing count.
Budget indicator on each card: given COP $1,766,000,000, show "Estimated properties purchasable: X" based on the market's average property price.
Sort cards by investment fit score by default. Allow re-sort by ADR, occupancy, or annual revenue.
Data from GET /markets via react-query.

Step 12 **Property deep-dive screen**
Route: /analyze
Input form (React state, not HTML form element): Finca Raiz URL or manual address, down payment % (default 30%), interest rate % (default 10%), HOA in COP (default 0 for rural, 350,000 for urban), management fee % (default 22%), renovation budget COP (optional).
On submit: POST to /properties then POST to /properties/{id}/analyze. Show skeleton loading cards during API calls.
Results panel: 9 metric cards in a 3×3 grid. ADR, occupancy, and projected revenue at top. Monthly expenses, annual net income, cash invested in the middle. CoC return and payback period at bottom — color-coded green (CoC > 8%), amber (5–8%), red (< 5%).
Revenue comparison row: AirROI direct estimate vs. comp-derived estimate side by side. Label the lower as "conservative".
Sensitivity table: 3×3 grid of CoC return at ADR -10%/base/+10% vs occupancy -10%/base/+10%. Computed in TypeScript from base figures — no API call.
Peak months: 12-bar Recharts BarChart showing relative occupancy by month from seasonality data.
Currency toggle: all monetary values switch between USD and COP on one click, using the rate from GET /exchange-rate.

Step 13 **Comparable listings panel**
Rendered below the financials on the deep-dive screen.
Sortable table: address, property type, bedrooms, ADR (USD), occupancy %, est. annual revenue (USD), distance from target property in km.
Default sort: annual revenue descending.
Each row links to the Airbnb listing URL where available.
Data from GET /properties/{id}/comps/cached (first call fetches live; subsequent calls use the cached version).
Show a "Refresh comps" button that triggers a fresh call to GET /properties/{id}/comps.

Step 14 **Deploy**
Backend: push to GitHub. Connect Railway to the repo — Railway auto-detects Python and runs uvicorn main:app. Add the Railway PostgreSQL plugin. Set nightly ingest scripts as Railway Cron Jobs.
Frontend: connect the same or a separate repo to Vercel. Set VITE_API_URL to the Railway backend URL in Vercel environment variables. Automatic deploy on push to main.

Test the full flow from the deployed URL: add a Salento listing, run the analysis, verify all 9 metrics render, confirm comps load and the currency toggle works.
Share the URL with the team. The system is now live for market research.

Part 4 — What to Do Today

Six tasks to complete before end of day. These establish the foundation everything else builds on.

#	Task	Acceptance criteria
1	AirROI account	Sign up at airroi.com. API key visible in dashboard. $10 in credits added.
2	Coverage check	API returns market data (ADR + occupancy populated) for all 8 markets. Calima and Pance alternatives identified if coverage is thin. Raw JSON saved locally.
3	PostgreSQL running	PostgreSQL 16 running locally or on Railway. PostGIS enabled. All 5 tables created with correct schema. GiST index on listings.location confirmed.
4	Markets ingested	ingest_markets.py runs without error. All 8 markets have rows in the markets table with ADR and occupancy values present.
5	Calc functions	calculations.py exists with all 6 functions. Manual test with known inputs returns correct values. Example: COP 500M Calima finca at 30% down, 10% rate, 15 years, 0 HOA = ~USD 1,540/month mortgage.
6	Finca Raiz scrape	Scraper returns correct price in COP from a live Salento or Calima listing URL. Converts to USD using frankfurter.app.

Part 5 — Tech Stack Reference

Layer	Technology	Notes
Data ingestion	Python 3.11 + httpx + BeautifulSoup	AirROI API calls + Finca Raiz scraper + exchange rate fetch. Use httpx async for concurrent market ingestion.
Job scheduling	Railway Cron (MVP) / Celery + Redis (scale)	Nightly market refresh at 3am Colombia time (UTC-5). On-demand scraping triggered by POST /properties calls.
Database	PostgreSQL 16 + PostGIS	PostGIS GEOGRAPHY type for radius-based comp queries. GiST index on listings.location. Use asyncpg driver for async FastAPI routes.
Backend	Python + FastAPI + Pydantic v2	Pydantic v2 for response schemas. SQLAlchemy 2.0 async ORM or raw asyncpg. Swagger UI auto-generated at /docs. CORS configured for the Vercel frontend URL.
Frontend	React 18 + TypeScript + Vite	react-query for data fetching and caching. recharts for financial charts. react-leaflet for the Colombia map. Strict TypeScript mode throughout.
Hosting	Railway (backend) + Vercel (frontend)	Railway: auto-deploy from GitHub, built-in PostgreSQL plugin, cron jobs. Vercel: auto-deploy, CDN, free tier sufficient for MVP.
Exchange rate	frankfurter.app (free, no API key)	GET https://api.frankfurter.app/latest?from=USD&to=COP. Refresh daily. Store in exchange_rates table to avoid repeat calls.

Layer	Technology	Notes
STR data	AirROI API	22 REST endpoints. Market summary, occupancy, seasonality, listing search (market, radius, polygon), revenue calculator, comparables, future rates. $0.01/call, pay-as-you-go.

Part 6 — Colombia Operating Assumptions

Use these as default values in the calculation engine. All are editable by the user in the property analysis form.

Input	Default value	Notes
Mortgage interest rate	**10.0%**	Midpoint of typical 9–12% COP-denominated loans for foreign buyers. Colombia policy rate: 9.25% as of Jan 2026.
Down payment	**30%**	Conservative assumption. Some banks accept 20% for STR investment properties.
Loan term	**15 years**	Standard for investment properties in Colombia.
Property management fee	**22% of gross revenue**	Typical range 18–25% for STR management in Colombia. Includes cleaning coordination.
Maintenance reserve	**1% of purchase price / year**	Standard industry reserve. Divide by 12 for monthly figure.
Closing costs	**2.75% of purchase price**	Notary fees, registration taxes, escritura. Typical range 2.5–3% in Colombia.
Predial (property tax)	**0.8% of assessed value / year**	Ranges 0.5–1.5% depending on city and property value. Urban properties (Bogota) toward the higher end.
HOA (administración) — urban apartments	**COP 350,000 / month**	COP 200K–1.2M range in urban buildings. Pre-populate based on city: Bogota/Medellin = COP 500K, Cali urban = COP 300K.
HOA — Calima / Pance / Salento fincas	**COP 0 / month**	Standalone fincas and rural houses have no HOA. This is a significant cost advantage over urban apartments.
RNT registration	**Required in all 8 markets**	All markets require Registro Nacional de Turismo registration under 2026 enforcement rules. Free to register at Confecamaras. One-time setup.
Currency display	**Both COP and USD**	Store all values in both currencies. Use a daily COP/USD rate refreshed from frankfurter.app. Display a toggle in the UI for switching between the two.

Part 7 — Estimated Running Costs

Item	Est. monthly cost	Notes
AirROI API — market ingestion (8 markets, nightly)	~$15–$30	8 markets × 30 days × ~$0.01/call for summary endpoints.

Item	Est. monthly cost	Notes
AirROI API — comps + property analysis (50/month)	~$10–$25	Comparables and revenue calculator calls per property analyzed.
Finca Raiz scraper	$0	Low volume, on-demand only. No proxies needed at this scale.
Railway (backend + database + cron)	$5–$20	Starter plan covers MVP. Includes PostgreSQL plugin and cron scheduler.
Vercel (frontend)	$0	Free tier is more than sufficient for this workload.
frankfurter.app (exchange rate)	$0	Free, no API key required.
Total MVP running cost	**~$30–$75 / month**	Compared to AirDNA at $300–$500/month for equivalent data coverage.

> **Next step after today:** Once the AirROI coverage check confirms usable data for Calima and Salento (Step 1), run the full data ingestion for those two markets first and complete an end-to-end financial analysis on 3–5 real property listings using COP $1,766,000,000 as the purchase ceiling. This produces the first real investment signal from the system within 3 days.